

CG3207 Computer Architecture

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About the Course

- All about how computers *really* work
- We will be ~~using~~ creating a processor!
- Lab Information & FAQs will be in Github- <https://nus-cg3207.github.io/labs/>
- Assessment
 - 60% CA
 - Lab assignments – 45%
 - Lecture quizzes – 15%
 - 40% Final Exam
 - MCQs / short questions
 - Printed and handwritten reference materials (no restriction on quantity of materials) allowed; no electronic references
 - Calculators allowed



Course Learning Outcomes

CLO1	Able to write a synthesizable HDL Program for a given digital logic circuit and implement it on a FPGA platform
CLO2	Able to design and develop a simple and functional Arithmetic & Logic Unit (ALU) of a given CPU using the given specifications on operations
CLO3	To implement a data-path and control unit for a processor and to execute a complete program using the given set of instructions for a given CPU
CLO4	Basic understanding of techniques for performance enhancement, multi-processor systems and associated issues in architecture, memory and I/O handling



Course Outline

- History, technology trends, performance
- Digital design using HDLs
- Basic processor design (datapath & controller)
- ALU design
- Pipelining, performance enhancement
- Memory system principles
- I/O principles
- Multiprocessor/core system concepts

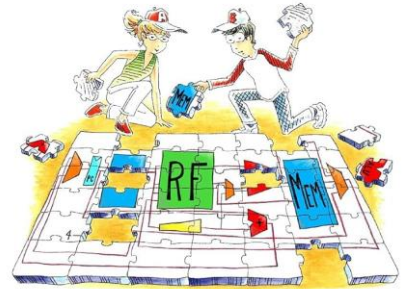
Books Used in this Course

Digital Design and Computer Architecture:
RISC-V Edition by Sarah Harris and David
Harris.
(the design I teach is my own, differs from the
book)

Computer Organization and Design RISC-V
Edition

PS : Lecture notes are meant to complement the
book, not to substitute it. Please read these classic
books!

Digital Design and Computer Architecture RISC-V Edition



MK
Morgan Kaufmann

Sarah L. Harris
David Harris

COMPUTER ORGANIZATION AND DESIGN THE HARDWARE/SOFTWARE INTERFACE RISC-V EDITION





(not so) Great Expectations

Revise and be ready on the following contents!

- Binary number system (signed and unsigned numbers), Assembly language, basic I/O interfaces
- Digital logic & arithmetic, flip-flops / registers, state-machines, VHDL/Verilog Programming – self-learning if you have not taken EE2026



Lab Assignments

Labs will be face to face. You will be given one FPGA board per team.

Assignment 1	VHDL/Verilog, ASM Exercise	10 marks
Assignment 2	Basic CPU Design	30 marks
Assignment 3	ALU Design	20 + 5@ marks
Assignment 4	Advanced CPU design	15 + 10@ marks

@: Performance improvement attempts.

Assignment 1: Individual assignment

Assignment 2 - 4: Work individually or in groups of up to 3 students from the **same lab day**. *There will be intra-team differentiation in marks.*



DOs and DON'Ts

- Avoid explicit emailing of codes & asking for debugging / spotting the errors
- You will need to use internet resources/data sheets, etc. as a part of your design journey
- Individual emails regarding lab assignments are **strongly discouraged**. Please post your doubts as a discussion on GitHub. If the matter is personal, please email (via Canvas Inbox, not normal email) the Lab TA, with a cc to me. Allow a couple of working days to respond to any email



DOs and DON'Ts ...

- Lecture notes, lab materials, lecture videos etc. are strictly for your own learning. DO NOT post / upload / link it anywhere
- Keep your code repositories **private**, even after you have completed the Course
- **NO Copying** from peers &/or from Internet. You learn much more when you do it yourself
You can Google.. so can we 😊

Let the fun begin ...

